**Supporting Information**

# Personalized Medication for Chronic Diseases using Multimodal Data-Driven Chain-of-Decisions

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**Appendix 1：Datasets**

Regarding the generalizability of the model between diseases and hospitals. The complete multimodal datasets for these diseases (CKD, MN, RA, CRC,) are rare datasets, and the collection of clinical characteristics and pathological mechanisms for each disease requires different devices and processes, making the collection and analysis of relevant data challenging. To the best of our knowledge, there currently exists no publicly available dataset worldwide that simultaneously incorporates multimodal clinical data and Chinese and Western medicine data for the chronic diseases at a comparable scale. The dataset we have constructed (CKD, MN, RA, CRC) fills this gap, aiding researchers in conducting rare disease research.

The inclusion and exclusion criteria for data collection are as follows: Inclusion criteria: ① Patients with a clear diagnosis and complete medical records of Chronic Kidney Disease (CKD), Rheumatoid Arthritis (RA), Colorectal Cancer (CRC); or Knee Osteoarthritis (KOA). ② The dataset must include at least one type of structured laboratory test or imaging information, as well as medication information. Exclusion criteria: ① Severe missing data or insufficient follow-up data; ② Patients with other major systemic comorbidities.

A renal biopsy of histopathological slices was performed by nephrologists from Guangdong Provincial Hospital of Traditional Chinese Medicine and Zhongshan City Hospital of Traditional Chinese Medicine for a renal puncture biopsy. Histopathological slices of renal biopsies were selected under PAS stain. Colorectal biopsy histopathological slides were taken by a specialist from Jinan University who performed a gross biopsy of the excised tissue and selected colorectal histopathological slides under hematoxylin-eosin (HE) staining. Hamamatsu's pathology digital slide scanner was used to capture the whole slide imaging (WSIs) of the sample slides, and all the contaminated and poorly stained slides were excluded and saved as '.ndpi' data format files. Rheumatoid arthritis joint ultrasound images were examined by ultrasound specialists of Guangdong Provincial Hospital of Traditional Chinese Medicine (GDHTC) and saved as DCM files, and Sante DICOM

Viewer Lite (v4.0.2) was used to export each image in the DCM files to a '.jpg' format file.

Table S1. Baseline characteristics of RA dataset

| Clinical Feature | Number of Cases | Disease Activity Score in 28 Joints: Remission | Low Disease Activity | Moderate Disease Activity | High Disease Activty | $\chi^2$ | *P* |
|---|---|---|---|---|---|---|---|
| **Age** | | | | | | 10.109 | 0.271 |
| <50 | 43(26.7) | 5(11.6) | 1(2.3) | 15(34.9) | 22(51.2) | | |
| 50~58 | 43(26.7) | 3(7.0) | 1(2.3) | 15(34.9) | 24(55.8) | | |
| 58~67 | 36(22.4) | 1(2.8) | 1(2.8) | 11(30.5) | 23(63.9) | | |
| >67 | 39(24.2) | 0(0) | 1(2.6) | 8(20.5) | 30(76.9) | | |
| **Gender** | | | | | | 1.601 | 0.65 |
| Male | 22(13.7) | 1(4.5) | 1(4.5) | 8(36.5) | 12(54.5) | | |
| Female | 139(86.3) | 8(5.8) | 3(2.2) | 41(29.5) | 87(62.6) | | |
| | | | | | | 5.43 | 0.796 |
| 0 | 38(23.6) | 1(2.6) | 1(2.6) | 12(31.6) | 24(63.2) | | |
| 1 | 63(39.1) | 3(4.8) | 1(1.6) | 19(30.2) | 40(63.5) | | |
| 2 | 45(28.0) | 4(8.9) | 2(4.4) | 11(24.4) | 28(62.2) | | |
| 3 | 15(9.3) | 1(6.6) | 0(0) | 7(46.7) | 7(46.7) | | |
| **Synovial blood flow** | | | | | | 4.735 | 0.881 |
| 0 | 45(28.0) | 3(6.7) | 1(2.2) | 12(26.7) | 29(64.4) | | |
| 1 | 53(32.9) | 3(5.7) | 1(1.9) | 13(24.5) | 36(67.9) | | |
| 2 | 43(26.7) | 2(4.7) | 2(4.7) | 16(37.2) | 23(53.5) | | |
| 3 | 20(12.4) | 1(5.0) | 0(0) | 8(40.0) | 11(55.0) | | |
| **Bone erosion** | | | | | | 1.23 | 0.76 |
| Yes | 140(87.0) | 9(6.4) | 4(2.9) | 43(30.7) | 84(60.0) | | |
| No | 21(13.0) | 0(0) | 0(0) | 6(28.6) | 15(71.4) | | |

As show in Table S1, age was non-normally distributed according to the Shapiro–Wilk test and was categorized based on quartiles (Q1=50, Q2=58, Q3=67 years). Group comparisons were performed using Pearson's chi-square test. Two-sided *P* values < 0.05 were considered statistically significant.

Table S2. Baseline characteristics of CKD dataset

| Clinical Feature | Number of Cases | CKD Stage: 1 | 2 | 3~5 | $\chi^2$ | *P* |
|---|---|---|---|---|---|---|
| **Age** | | | | | 9.07 | 0.059 |

| | | | | | | |
|---|---|---|---|---|---|---|
| <30 | 26(23.9) | 20(76.9) | 2(7.7) | 4(15.4) | | |
| 30~50 | 61(56.0) | 37(60.7) | 16(26.2) | 8(13.1) | | |
| ≥50 | 22(20.1) | 9(40.9) | 6(27.3) | 7(31.8) | | |
| **Gender** | | | | | 0.051 | 0.975 |
| Male | 57(52.3) | 34(59.7) | 13(22.8) | 10(17.5) | | |
| Female | 52(47.7) | 32(61.5) | 11(21.2) | 9(17.3) | | |
| **Pathological Lesion** | | | | | 31.691 | 0.000 |
| Mild Lesion | 48(44.0) | 41(85.4) | 6(12.5) | 1(2.1) | | |
| Moderate Lesion | 48(44.0) | 23(47.9) | 14(29.2) | 11(22.9) | | |
| Severe Lesion | 13(12.0) | 2(15.4) | 4(30.8) | 7(53.8) | | |

As show in Table S2, age followed a normal distribution as assessed by the Shapiro–Wilk test (P = 0.107). The mean ± standard deviation of age was 40.83 ± 12.55 years. Age was categorized based on clinically relevant cutoffs into three groups: young (<30 years), middle-aged (30–50 years), and elderly (≥50 years). Pathological lesion was assessed according to Katafuchi Score.

Table S3. Baseline characteristics of MN dataset

| Clinical Feature | Number of Cases | Ehrenreich-Churg Stage | | | $\chi^2$ | *P* |
|---|---|---|---|---|---|---|
| | | **I** | **II** | **III** | | |
| **Age** | | | | | 3.035 | 0.899 |
| <34 | 20(27.8) | 1(5.0) | 18(90.0) | 1(5.0) | | |
| 34~50 | 16(22.2) | 1(6.3) | 15(93.8) | 0(0) | | |
| 50~60 | 19(26.4) | 3(15.8) | 15(78.9) | 1(5.3) | | |
| >60 | 17(23.6) | 2(11.8) | 14(82.4) | 1(5.9) | | |
| **Gender** | | | | | 0.799 | 0.851 |
| Male | 49(68.1) | 4(8.2) | 43(87.8) | 2(4.1) | | |
| Female | 23(31.9) | 3(13) | 19(82.6) | 1(4.3) | | |
| **Risk Assessment** | | | | | 1.921 | 0.726 |
| Low Risk | 5(6.9) | 1(20.0) | 4(80.0) | 0(0) | | |
| Moderate Risk | 28(38.9) | 3(10.7) | 24(85.7) | 1(3.6) | | |
| High Risk | 39(54.2) | 3(7.7) | 34(87.2) | 2(5.1) | | |

As show in Table S3, age was non-normally distributed according to the Shapiro–Wilk test and was categorized based on quartiles (Q1=34, Q2=49.5, Q3=60 years). Group comparisons were performed using Pearson's chi-square test. Two-sided *P* values < 0.05 were considered statistically significant. Ehrenreich–Churg staging was used for membranous nephropathy classification based on electron microscopy: Stage I = small subepithelial electron-dense deposits without GBM reaction; Stage II = deposits surrounded by projections of GBM forming "spikes" or new basement membrane material; Stage III = deposits incorporated within thickened

and irregular GBM; Stage IV (not included here) = resolution of deposits with persistent GBM alterations. Risk categories (low, moderate, and high) were determined based on clinical criteria outlined in the 2021 KDIGO guideline for glomerular diseases, considering parameters such as eGFR, proteinuria, serum albumin, and anti-PLA2R antibody levels.

Table S4. Baseline characteristics of CRC dataset

| Clinical Feature | Number of Cases | Degree of stenosis | | $\chi^2$ | P |
|---|---|---|---|---|---|
| | | <50% | >50% | | |
| **Age** | | | | 2.238 | 0.536 |
| <57 | 43(26.9) | 27(62.8) | 16(37.2) | | |
| 57~69 | 41(25.6) | 23(56.1) | 18(43.9) | | |
| 69~78 | 41(25.6) | 26(63.4) | 15(36.6) | | |
| >78 | 35(21.9) | 17(48.6) | 18(51.4) | | |
| **Gender** | | | | 1.003 | 0.317 |
| Male | 105(65.6) | 64(61.0) | 41(39.0) | | |
| Female | 55(34.4) | 29(52.7) | 26(47.3) | | |
| **TNM Stage** | | | | 5.375 | 0.068 |
| I | 44(27.5) | 32(72.7) | 12(27.3) | | |
| II | 98(61.2) | 52(53.1) | 46(46.9) | | |
| III | 18(11.3) | 9(50.0) | 9(50.0) | | |
| Degree of differentiation | | | | 8.451 | 0.038 |
| Well differentiated | 24(15.0) | 20(83.3) | 4(16.7) | | |
| Moderately differentiated | 62(38.8) | 31(50.0) | 31(50.0) | | |
| Poorly differentiated | 56(35.0) | 33(58.9) | 23(41.1) | | |
| Undifferentiated | 18(11.2) | 9(50.0) | 9(50.0) | | |
| **Survival Time** | | | | 1.496 | 0.221 |
| Alive | 49(30.6) | 32(65.3) | 17(34.7) | | |
| Deceased | 111(69.4) | 61(55.0) | 50(45.0) | | |

As show in Table S1, age was non-normally distributed according to the Shapiro–Wilk test and was categorized based on quartiles (Q1=57, Q2=69, Q3=78 years). Group comparisons were performed using Pearson's chi-square test. Two-sided *P* values < 0.05 were considered statistically significant. "*Degree of stenosis"* was determined based on the estimated luminal narrowing rate: <50% indicates mild to moderate stenosis, and >50% indicates severe stenosis. TNM stage (Tumor–Node–Metastasis classification) was determined according to the 8th edition AJCC criteria: Stage I (T1–T2, N0, M0), Stage II (T3–T4, N0, M0), and Stage III (any T, N1–2, M0); Stage 0 and IV were not included in this dataset. Tumor

differentiation was graded histologically as well, moderately, poorly, or undifferentiated, based on the degree to which tumor cells resemble normal glandular epithelium.

Table S5. Baseline characteristics of KO dataset

| Clinical Feature | Number of Cases | Kellgren-Lawrence grading system | | | | $\chi^2$ | P |
|---|---|---|---|---|---|---|---|
| | | 0 | I | II | III | | |
| **Age** | | | | | | 12.897 | 0.167 |
| <60 | 794(25.0) | 197(24.8) | 315(39.7) | 250(31.5) | 32(4.0) | | |
| 60~67 | 895(28.2) | 185(20.7) | 375(41.9) | 278(31.1) | 57(6.4) | | |
| 67~74 | 714(22.5) | 177(24.8) | 295(41.3) | 204(28.6) | 38(5.3) | | |
| >74 | 770(24.3) | 176(22.9) | 305(39.6) | 236(30.6) | 53(6.9) | | |
| **Gender** | | | | | | 5.876 | 0.118 |
| Male | 849(26.8) | 218(25.7) | 338(39.8) | 254(29.9) | 39(4.6) | | |
| Female | 2324(73.2) | 517(22.2) | 952(41) | 714(30.7) | 141(6.1) | | |

As show in Table S5, age was non-normally distributed according to the Shapiro–Wilk test and was categorized based on quartiles (Q1=60, Q2=67, Q3=74 years). Group comparisons were performed using Pearson's chi-square test. Two-sided *P* values < 0.05 were considered statistically significant. The Kellgren–Lawrence grading system was used to assess radiographic severity of osteoarthritis: grade 0 = no radiographic features of OA, grade I = doubtful narrowing of joint space and possible osteophytic lipping, grade II = definite osteophytes and possible joint space narrowing, grade III = multiple osteophytes, definite narrowing of joint space, and possible deformity of bone ends.

## Appendix 2：MDD-CoD Model Architecture Diagram

The implementation code for the model proposed in this study has been open-sourced on GitHub and is publicly available. The repository URL is: https://github.com/yhengYE/MDD-CoD. All configuration details, preprocessing scripts, and model architecture definitions required for complete experiment reproduction are included in this repository. The complete dataset is subject to privacy and ethical restrictions (IRB: BE2025-098-01, BE2024-325-01, 2024ZSZY-LLK-408, and JNUKY-2023-0105) and cannot be publicly shared. To obtain the dataset, please contact the corresponding author and sign the relevant confidentiality agreement before accessing it.

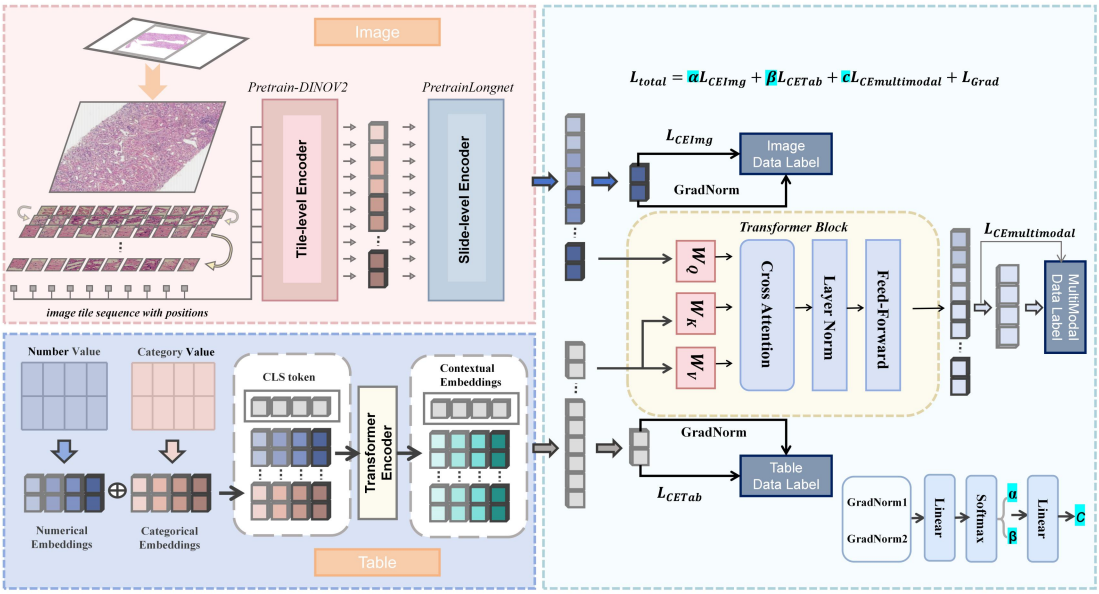


Figure S1. MDD-CoD1： The multi-modal feature embedding and multi-task prediction model based on Transformer improves the definition of disease classification.

As shown in Figure S1, a multi-modal fusion framework based on the multi-task thought is proposed, which regards the feature representation process of each mode as a task, and the alignment and fusion of multi-modal features as a global task. This framework can be applied to downstream tasks that address the weight of different modes in clinical decision making. Specifically, each single mode carries out feature learning through its own pre-trained feature extractor, and then carries out information

exchange between modes through the cross-attention mechanism introduced by the global task. Finally, the common loss of single mode and multi-mode is synthesized, the loss optimization of the overall model is carried out, and comprehensive clinical decision is obtained, and the proportion of different clinical modes in decision making is obtained.

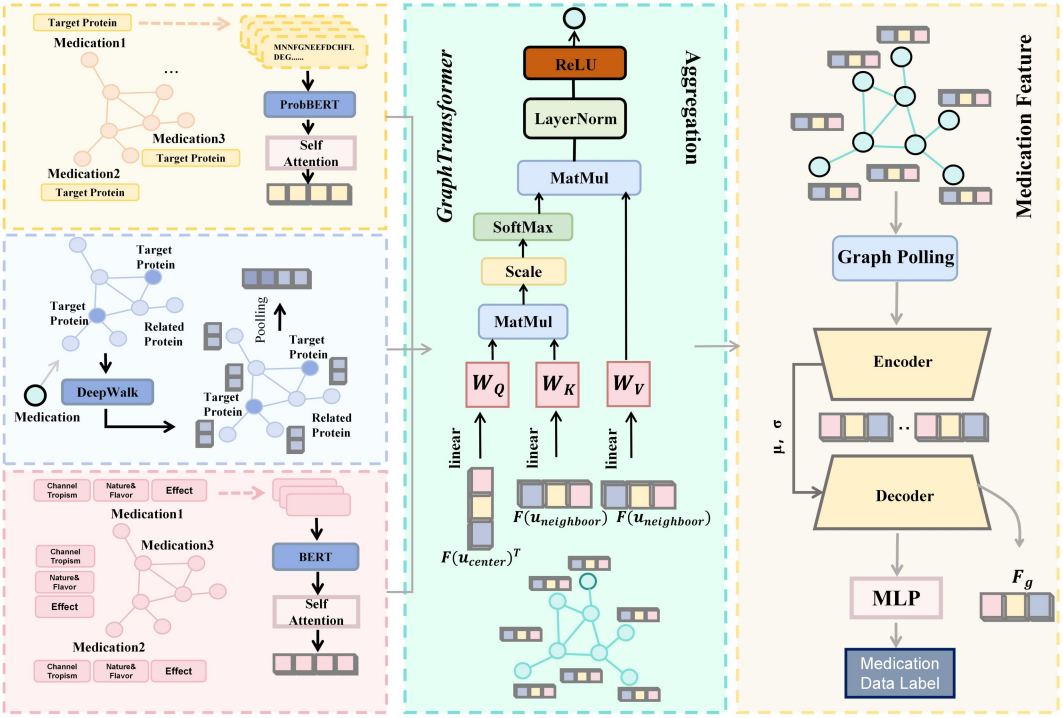


Figure S2. MDD-CoD2: Graph transformer network-based drug marco-micro information fusion model to analyze the therapeutic mechanism of drug combination.

As shown in Figure S2, a graph neural network based on the three modalities of drug information is constructed. On the one hand, for macro pharmacological modal information, BERT-Chinese pre-training model is used to embed text features. In addition, a fusion module based on attention mechanism is introduced to merge pharmacological information of multiple dimensions to obtain a unified feature representation. This feature can represent the macro information of the drug. On the other hand, for microscopic modal information, ProbBERT pre-training model is used to extract drug target characteristics. The pre-training model has been pre-trained on large-scale protein sequence data sets, and can better understand the meaning of protein amino acid sequence and give high-quality feature representation. Then we enter the attentional mechanism and fuse multiple protein sequences of a single drug

to obtain a uniform representation of protein characteristics. This feature can represent the microscopic protein information of the drug. In addition, the micro-characterization also incorporates a protein-protein interaction network (PPI) of the drug target based on DeepWalk, aiming to build a comprehensive representation of the PPI characteristics.

Macro and micro drug prescription information features (including macro drug, micro protein properties, and micro protein network structure) are prepared for splicing operation, and a unified macro and micro drug prescription feature representation was obtained. Then, a drug prescription is regarded as a subnetwork, a drug is a network node, and macro and micro drug prescription features are node features. The Gnn-Transformer model is used to reconstruct and aggregate the network to obtain the macro and micro representation of drug prescription network after fusion based on GNN. After the pooling operation, the macro and micro drug prescription network obtains the prescription feature representation, which is sent to the VAE module for the alignment of multi-modal data. The input features are compressed into a hidden space representation, and the macro and micro drug prescription feature representations are reconstructed.

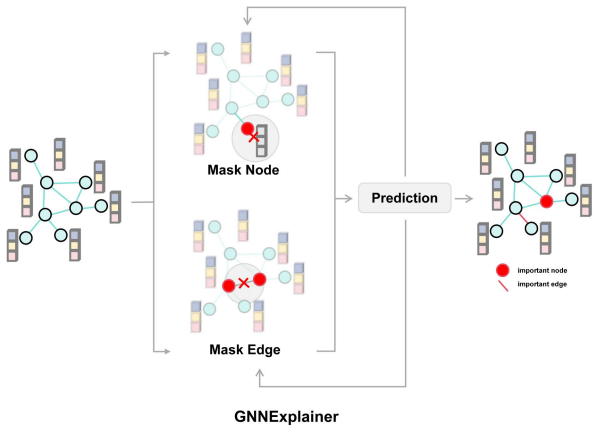


Figure S3. Interpretability of the MDD-CoD2 model.

As shown in Figure S3, Graph neural network interpretable tool GNN-explainer is used to get the clinical hints in Chain2. GNN-explainer, through a trainable masking mechanism, dynamically identifies the most critical substructures of the network for a specific prediction. Specifically, it assigns importance scores to the

node and edge features in the drug network and, through an optimization process, finds a concise subgraph that can reproduce the original prediction results to the greatest extent in the GNN model.

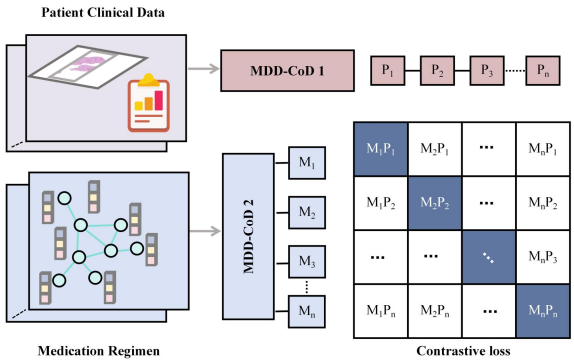


Figure S4. MDD-CoD3：A cross-modal shared space representation model based on contrast learning for disease and medication regimen matching.

The above two chain prepare the patient's multi-modal clinical features and macro-micro medication prescription features, and the stage3 (as shown in Figure S4) builds CLIP model based on these two types of features. Specifically, multimodal clinical features and macro- micro drug prescription features are preprocessed and fed into the two inputs of the CLIP model, respectively, to embed into a shared semantic space for constructing positive and negative sample pairs. Then, through the InfoNCE loss optimization model of comparative learning, clinical features and prescription features are effectively aligned in semantic space, and accurate correlation modeling of the two types of features is realized.

## Appendix 3 ： Training Details

**MDD-CoD1:** To conserve computational resources, we first performed standalone feature extraction for the imaging modality of patients requiring subsequent training, and saved the extracted features as matrix files. As a result, the imaging modality model did not participate in parameter updates during subsequent training. Since the imaging features were derived using a pre-trained model and this model was not integrated into the full pipeline, no further parameter tuning was conducted for it.

To accommodate this design, the MDD-CoD1 model underwent full-parameter pre-training of the FT-Transformer, during which all subsequent layers—including the unimodal classification layer, multimodal fusion layer, and multimodal classification layer—were temporarily frozen. After the FT-Transformer pre-training phase, all of its layers were permanently frozen, and only the parameters of the unimodal classification layer, multimodal fusion layer, and final classification layer were updated during the training phase. The key hyperparameters are listed in the table below.

We employed a random split strategy for dataset partitioning, allocating 70% of the data to the training set and 30% to the test set. Detailed training hyperparameters are provided in the table below.

Table S6. MDD-CoD1 Hyperparameter

| **Hyperparameter** | **RA** | **MN** | **CRC** | **CKD** |
|---|---|---|---|---|
| Input dim | 768 | 768 | 1536 | 192 |
| Epochs | 400 | 300 | 300 | 200 |
| Pretrain epochs | 100 | 70 | 70 | 30 |
| Optimizer | AdamW | AdamW | AdamW | AdamW |
| Learning rate | 1e-4 | 1e-3 | 1e-3 | 1e-4 |
| Decay weight | 2e-3 | 1e-4 | 5e-4 | 1e-4 |

**MDD-CoD2：** For the MDD-CoD2 model, we adopted 4-fold cross-validation (K=4), dividing the original dataset into four subsets. In each iteration, three subsets were used for training and one for validation, with each subset serving as the validation set exactly once. The main evaluation metrics included AUROC, F1-score,

and Precision. Key hyperparameters are also summarized in the table.

Table S7. MDD-CoD2 Hyperparameter

| **Hyperparameter** | **RA** | **MN** | **CKD** |
|---|---|---|---|
| Batch size | 256 | 256 | 256 |
| Epochs | 1000 | 300 | 1000 |
| Optimizer | AdamW | AdamW | AdamW |
| Learning rate | 1e-3 | 1e-3 | 1e-4 |
| Decay weight | 2e-3 | 2e-3 | 1e-4 |
| Scheduler | Steplr（70\0.85） | Steplr（50\0.85） | Steplr（70\0.85） |
| Memory using | 7GB | 7GB | 7GB |
| Training time | 6.68h | 1.33h | 6h |
| OverSampler | False | True | False |

**MDD-CoD3:** Prior to training the MDD-CoD2 model, patient and medication features were comprehensively extracted using the preceding two-stage models and saved as matrix files. During model development, the dataset was randomly partitioned into 70% for training and 30% for testing. The specific training hyperparameters are listed in the table below.

Table S8. MDD-CoD3 Hyperparameter

| **Hyperparameter** | **value** |
|---|---|
| Batch size | 128 |
| Input dim | 256 |
| hidden dim | 512 |
| output dim | 128 |
| Epochs | 1000 |
| Optimizer | Adam |
| Learning rate | 1e-4 |
| Decay weight | 2e-4 |

Given the current version of the dataset, we found it unnecessary to use an additional validation set for model selection. This is due to the limited dataset size—mainly resulting from the difficulty of collecting sufficient data for rare or complex diseases. Allocating a separate validation set would reduce the size and diversity of the training set, potentially degrading model performance. Our dataset construction approach helps bridge the current data gap in this field and lays a solid foundation for future deployment and scalability.

**Appendix 4 ： Evaluation of Indicators**

The evaluation metrics employed in this study are standard indicators commonly used in deep learning-based prediction tasks. The corresponding formulas are provided below.

In classification model evaluation, True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) are fundamental performance indicators used to describe prediction accuracy and error types. Specifically, TP and TN represent the number of correctly predicted positive and negative samples, respectively, while FP and FN denote the number of incorrectly predicted positive and negative samples, respectively.

For the MDD-CoD1 model, the evaluation metrics included the Area Under the Receiver Operating Characteristic Curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC). AUROC is a broadly applicable metric across most prediction tasks, while AUPRC is particularly sensitive to class imbalance. Their combined use helps mitigate potential biases introduced by using a single metric under imbalanced conditions.

For the MDD-CoD2 model, we employed AUROC, F1-Score, and Precision. The F1-Score compensates for the limitations of AUROC and Precision when evaluating models on imbalanced datasets, with a particular focus on the model’ s ability to correctly classify minority classes.

For the MDD-CoD3 model, we used Hit@K as the evaluation metric. Hit@K is widely used in recommendation systems and information retrieval, and measures whether the ground-truth item appears within the top-K predicted items. The corresponding formulas are as follows:

$$TPR=\frac{TP}{TP+FN} \quad (1)$$

$$FPR=\frac{TP}{TP+FN} \quad (2)$$

$$AUROC=\frac{1}{2}\sum_{i=1}^{n}(FPR_i\text{-}FPR_{i-1})(TPR_i\text{-}TPR_{i-1}) \quad (3)$$

$$P=\frac{TP}{TP+FP} \quad (4)$$

$$R=\frac{TP}{TP+FN} \quad (5)$$

$$AURRC=\frac{1}{2}\sum_{i=1}^{n}(R_i-R_{i-1})(P_i-P_{i-1}) \quad (6)$$

$$F1=\frac{2PR}{P+R}=\frac{2TP}{2TP+FP+FN} \quad (7)$$

For a single test sample, Hit@k is a binary metric: (8)

$$Hit@k=\begin{cases}1, \text{if the correct answer is among the top- k predictions}\\ 0, \text{otherwise}\end{cases}$$

The average Hit@k over a test set of N samples is computed as: (9)

$$Hit@k=\frac{1}{N}\sum_{i=1}^{N}Hit@k$$

## Appendix 5：Comparative Advantages of Core Technologies in MDD-CoD

**FT-Transformer's Superiority in MDD-CoD1.** In the MDD-CoD1 model, for the medical diagnostic form data, we introduce FT-Transformer as an embedding encoder for laboratory indicators, aiming to leverage its global dependency modelling capability across feature columns. Taking the RA dataset as an example, the category-based features are pain, limitation of movement, stiffness, and swelling of joints; and the numerical features are CRP, ESR, RF, AST, and ALT. A priori, there is a strong relationship between symptoms such as joint pain, limitation of movement, stiffness, and swelling, and inflammatory indicators such as CRP and ESR, in which the stronger the inflammatory response, the more severe the patient's symptoms. Liver indicators such as rheumatoid factor (RF) and AST/ALT can reflect the activity or complications of rheumatic diseases. The joint modeling of these features through the embedding layer can fully reflect the clinical characteristics of patients.

**Multi-attribute medication feature's Superiority in MDD-CoD2.** Macro-pharmacological characteristics (Properties) of medications include the typical representative description of the text of the medications, specifically the Chinese medicine natures, tastes, and meridian tropisms, as well as the side effects of Western medicines to reflect their macro-pharmacological effects in the field of traditional Chinese medicine. Meridian tropisms refer to the effect of the medication on the five zang-organs and the six fu-organs, which builds up the scope of application and efficacy of the medication. Natures and tastes reflect the nature of the medication and the characteristics of its action. The efficacy corresponds to the therapeutic symptoms. This will serve as the more intuitive functional information of the medications. As the macroscopic pharmacological characteristics of western medications, the side effects of western medications stem from the interaction of their chemical components with the human body. The selection of this as a pharmacological feature of western medications in this study aims to portray the concept of the combination of Chinese and western medications to reduce toxicity. The medications' microscopic protein

network features (PPI) include the protein network composed of the drug target and its related proteins, aiming to construct a comprehensive protein network structural feature representation, focusing on the interaction network topology of the medications' target proteins in the biological system to analyze the target protein's pivot, modularity, and functional conduction pathway in the global protein interaction network. Medication micro-protein sequence characterization (Protein-Seq) contains the medication's target protein sequences that decode features of protein structure and function.

**DeepWalk' s Superiority in Structure-Only PPI Network Embedding.** In graph neural networks (GNNs), the absence of node features reduces neighborhood aggregation to mere degree distribution statistics, preventing the extraction of high-order semantic information and thereby diminishing the advantages typically offered by GNNs.

Metapath2Vec, which is specifically designed for heterogeneous graphs, relies on predefined metapaths (e.g., "protein-function-protein") to guide random walks. However, in this task, the existence of only a single edge type causes these metapaths to degenerate into simple repeated edge traversals, which ultimately leads to suboptimal performance.

LINE preserves local structural information by using a degree-based negative sampling strategy to optimize first-order (direct neighbor) and second-order (shared neighbor) node similarities. However, it tends to treat high-degree nodes as hubs or structural centers while overlooking the importance of critical bridge proteins, and thus fails to capture the global topological characteristics of protein-protein interaction (PPI) networks.

This study conducts comparative experiments among DeepWalk, Graph Neural Networks, Metapath2Vec, and LINE for network embedding. The experimental results demonstrate that DeepWalk achieves the best performance in embedding PPI features. The strength of DeepWalk lies in its strong compatibility with the homogeneous structure of PPI networks. The node co-occurrence distributions derived from random walks effectively encode the network's topology: short walks

are sufficient to cover most nodes while capturing both hub nodes and crucial bridge nodes, thus yielding superior topological representations.

**Appendix 6: Model Generalizability Between Diseases and Hospitals**

To demonstrate the model's (MDD-CoD1) generalizability, we pre-trained it using data from one disease and then fine-tuned it using data from another disease (25% of the total dataset) to validate its cross-disease transferability. As shown in Table S9, in most transfer scenarios, MDD-CoD1 outperformed baseline models trained on the full target-domain data. Notably, in the transfer from colorectal cancer (CRC) to chronic kidney disease (CKD), both of which share similar image modalities, MDD-CoD1 achieved up to a 15% improvement in AUROC. Even in a less favorable setting, such as CRC to rheumatoid arthritis (RA), the performance gap was kept within 3%. These results demonstrate the strong generalization ability of MDD-CoD1, with limited target-domain data, the fine-tuned pretrained model can surpass baseline models trained from scratch on full data. Most pre-trained models achieve performance surpassing the benchmark model trained on the full dataset of the target disease after fine-tuning with a small portion (25%) of the target disease data. Additionally, we provide the loss curve of MDD-CoD1 on the training set to explain the underlying mechanism of this improvement. As shown in Figure S5, across four datasets, the loss curves (e.g., the comparison between red and blue lines) exhibit lower training loss and faster convergence rates, indicating that the knowledge learned by MDD-CoD1 on source domain data facilitates learning for downstream tasks, thereby validating the model's generalizability.

Regarding the generalizability across hospitals and different diseases, we agree that further hierarchical analysis is needed, along with external validation across different hospitals and datasets. In fact, our data already covers multiple hospitals. Additionally, we plan to further expand and maintain this dataset in future research.

Table S9. Model performance comparison between transfer learning and full fine-tuning approaches.

| （Source->Target） | AUROC↑ | AUPRC↑ |
|---|---|---|

| | | |
|---|---|---|
| CRC->RA | 0.9244（0.8989） | 0.8137（0.7481） |
| CRC->CKD | 0.6889（**0.7920**） | 0.5630（**0.5708**） |
| MN->CRC | 0.7231（**0.7736**） | 0.8977（**0.9164**） |
| CKD->MN | 0.7136（**0.7573**） | 0.5940（**0.6650**） |

Where “Source” refers to models trained on the complete dataset of the source disease, while “Target” refers to models trained based on the source disease. X(Y) denotes the test set accuracy on the original dataset (test set accuracy after small-sample transfer learning).

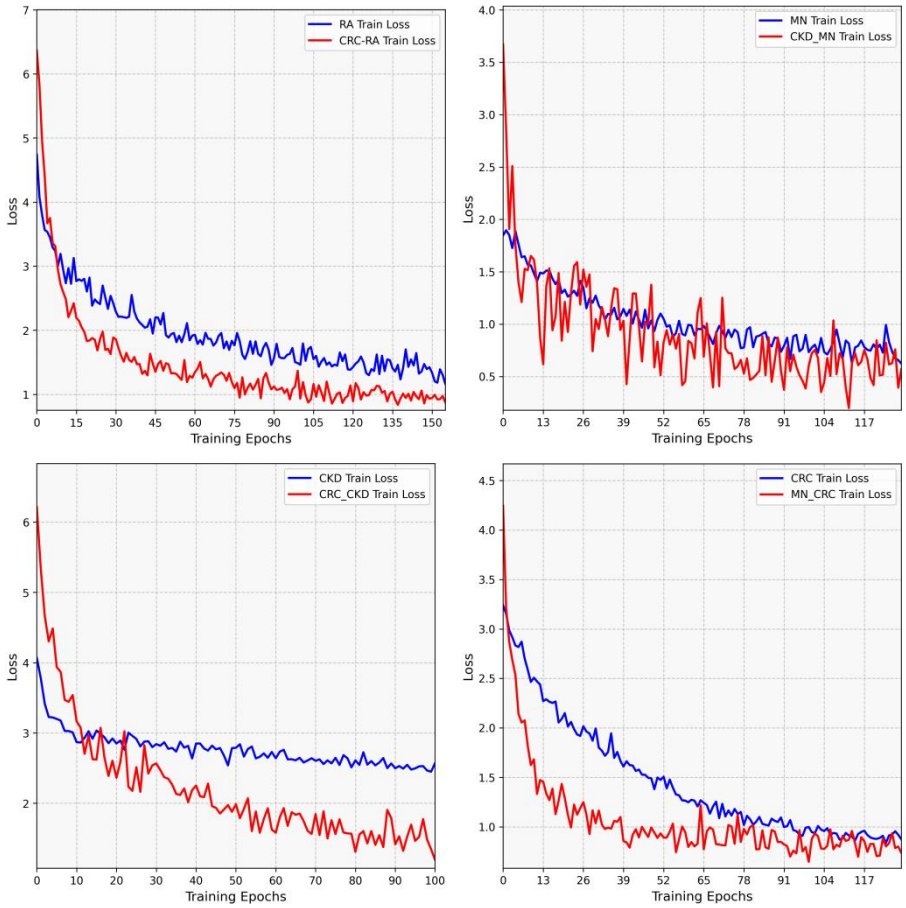


Figure S5: MDD-CoD demonstrates excellent generalization performance on multiple datasets. The blue curve represents the original training loss curve, while the red curve represents the training loss curve after small-sample transfer learning. On all four datasets, MDD-CoD demonstrates lower training loss and faster convergence speed.

## Appendix 7: MDD-CoD1 Interpretability Study

We used cosine similarity, a standard metric for cross-modal alignment research, to construct a modal feature interaction heatmap for interpretability research. By analyzing 13 random training samples from two datasets (CRC and MN), as shown in Figure S6: During the initial training phase, cross-modal similarity was generally low (CRC: -0.0363, MN: -0.0772). After training, the multi-modal features of both datasets showed an increase in average similarity (CRC: +0.1285, MN: +0.1602). This change confirms that the model has successfully established foundational semantic association capabilities, effectively validating the efficacy of MDD-CoD in cross-modal alignment tasks.

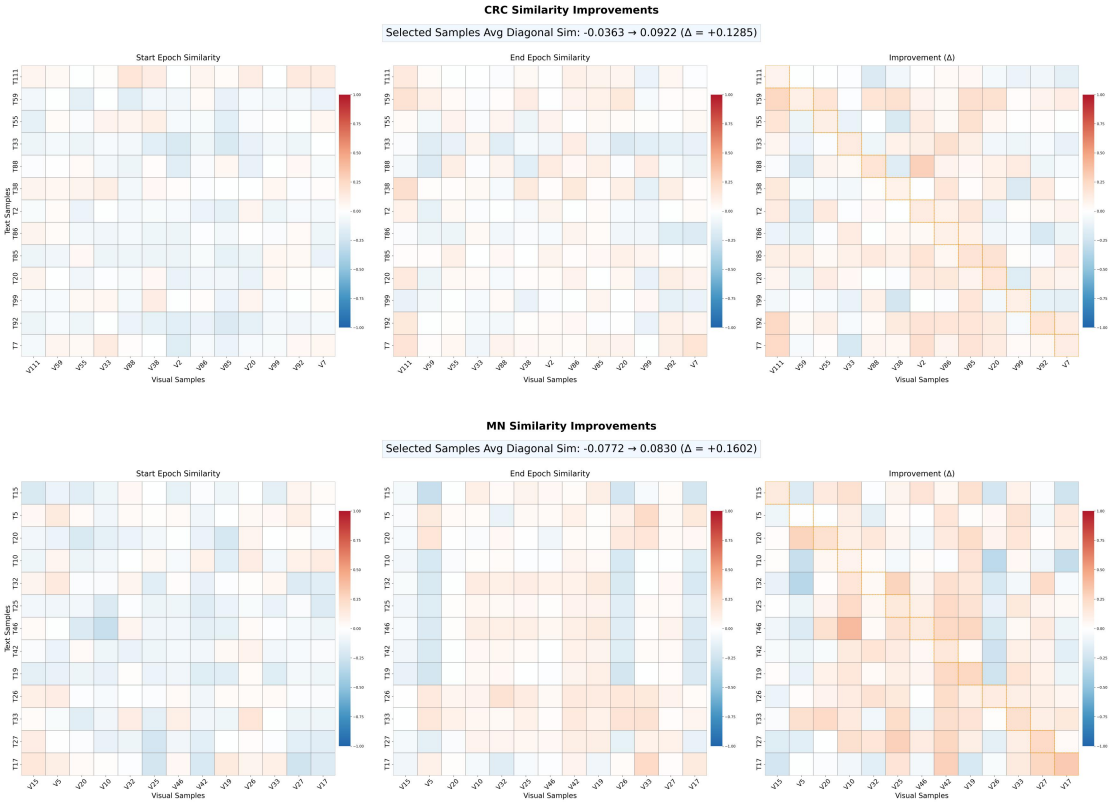


Figure S6: Cross-modal feature alignment heatmap on the CRC and MN datasets. A modal interaction heatmap based on cosine similarity, showing the evolution of cross-modal alignment from initial training (left) to final convergence (right) for 13 random samples. Blue tones (-0.1 to 0) indicate weak modal correlation, red tones (0

to 0.2) indicate strong cross-modal alignment.

## Appendix 8: Abbreviation of Medication

| Medication | Abbreviation of Medication |
|---|---|
| Compound Nephritis Tablets | CNT |
| Zingiberis Rhizoma Recens | Rhiz. Zing |
| Centellae Herba | C. Herba |
| Jujubae Fructus | J. Fructus |
| Lycii Fructus | L. Fructus |
| Rubiae Radix Et Rhizoma | R. Radix |
| Aspirin | Rad. Rub. |
| Cordyceps cicadae | C. cicadae |
| Aucklandiae Radix | A. Radix |
| Terazosin | TER |
| Platycodonis Radix | P. Radix |
| Poriae Cutis | P. Cutis |
| Rhapontici Radix | R. Radix |
| Irbesartan | IRB |
| Dianthi Herba | D. Herba |
| Pitavastatin | PTV |
| Lophatheri Herba | L. Herba |
| Amlodipine Besilate | AML-B |
| Rosuvastatin | Rosuv. |
| Cyperi Rhizoma | Rhiz. Cyperi |
| Dioscoreae Hypoglaucae Rhizoma | Rhiz. Diosc. Hypogl. |
| Microcos Paniculata | Microcos |
| Poria | Por. |
| Pheretima | Pher. |
| Hemocoagulase Injection | Inj. Hemocoag |
| Forsythiae Fructus | Fruct. Forsyth. |
| Valsartan And Amlodipine Tablets A | Tab. Vals. & Amlod. |
| Bambusae Caulis In Taenias | Caul. Bamb. |
| Pogostemonis Herba | Herb. Pogost. |
| Salviae Miltiorrhizae Radix Et Rhizoma | Rad. et Rhiz. Salviae |
| Sanqi Oral Liquid | Liq. Oral. Notogins |
| Jinshuibaopian | JSB |
| Paeoniaeradix Rubra | Rad. Paeon. Rub. |
| Polygoni Multiflori Caulis | Caul. Polyg. Mult. |
| Aluminum Magnesium Carbonate And Aspirin | Alum. Mag. Carb. & Asp. |
| Anemarrhenae Rhizoma | Rhiz. Anemarrh. |

| | |
|---|---|
| Scutellariae Radix | Rad. Scutell. |
| Terazosin Hydrochloride | Teraz. HCl |
| Fritillariae Thunbergii Bulbus | Bulb. Fritill. Thunb |
| Euryales Semen | Sem. Euryal. |
| Imperatae Rhizoma | Rhiz. Imper. |
| Clematidis Radix Et Rhizoma | Rad. et Rhiz. Clemat. |
| Pitavastatin | Pitav. |
| Chuanxiong Rhizoma | Rhiz. Chuanx |
| Alismatis Rhizoma | Rhiz. Alism. |
| Polyporus | Polypor. |
| Polyene Phosphatidylcholine | Phosph. Polyen. |
| Semen Armeniacae Amarum | Sem. Armen. Am. |
| Glycyrrhizae Radix Et Rhizoma Praeparata Cum Melle | Rad. et Rhiz. Glycyrrh. Praep. |
| Zingiberis Rhizoma | Rhiz. Zingib. |
| Scleromitrion Diffusum | Scleromit. |
| Methylprednisolone | Methylpred. |
| Nelumbinis Stamen | Stam. Nelumb. |
| Carvedilol | Carved. |
| Torsemide | Torsem. |
| Corni Fructus | Fruct. Corni |
| Cyclophosphamide | Cyclophosph. |
| Dendrobium With Fruits | Dendrob. c. Fruct. |
| Bupleuri Radix | Rad. Bupl. |
| Fluvastatin | Fluvast. |
| Corni Fructus | Fruct. Corni |
| Cuscutae Semen | Sem. Cusc. |
| Huangkui Jiaonang | HKJN |
| Atractylodis Macrocephalae Rhizoma | Rhiz. Atract. Macro. |
| Recombinant Human Erythropoietin Polysaccharide-Iron Complex Capsules | Caps. EPO-PIC |
| Complexus Polysaccharidi-Ferrosus | CPF |
| Low Molecular Weight Heparin | Heparin. LMWH |
| Furosemide Tablets | Tab. Furos. |
| Compound Ketoacid Tablets | Tab. Ketoacid. Co. |
| Eucommiae Cortex Praeparatus | Cort. Eucomm. Praep. |
| Low Molecular Weight Heparin Calcium | Heparin. Calc. LMWH |
| Bisoprolol Fumarate | Bisoprol. Fum. |
| Calcitriol | Calcitriol. |
| Gliclazide | Gliclaz. |
| Acarbose | Acarb. |
| Albiziae Flos | Fl. Albiz. |
| Codonopsis Radix | Rad. Codon. |

| | |
|---|---|
| Spironolactone | Spironol. |
| Calcium Carbonate D3 | Calc. Carb. D3 |
| Asari Radix Et Rhizoma | Rad. et Rhiz. Asari |
| Talcum | Talc. |
| Calcium Carbonate D3 | Calc. Carb. D3 |
| Gentianae Macrophyllae Radix | Rad. Gent. Macroph |
| Chaenomelis Fructus | Fruct. Chaenom. |
| Cinnamomi Ramulus | Ram. Cinnam. |
| Fluvastatin | Fluvast. |
| Atractylodis Rhizoma | Rhiz. Atract |
| Diospyros Iotus Linn | Diosp. Lot. |
| Glycyrrhizae Radix Et Rhizoma | Rad. et Rhiz. Glycyrrh. |
| Polygoni Cuspidati Rhizoma Et Radix | Rhiz. et Rad. Polyg. Cusp. |
| Atractylodis Macrocephalae Rhizoma | Rhiz. Atract. Macro. |
| Raphani Semen | Sem. Raph. |
| Oryzae Fructus Germinatus | Fruct. Oryz. Germ. |
| Asini Corii Colla | Coll. Cor. Asin. |
| Ostreae Concha | Conch. Ostr. |
| Schisandrae Chinensis Fructus | Fruct. Schisand. Chin. |
| Pyrrosiae Folium | Fol. Pyrros. |
| Uremic Clearance Granule | Gran. Urem. Clear. |
| Tacrolimus | Tacrol. |
| Lonicerae Japonicae Flos | Fl. Lonic. Jap. |
| Coicis Semen Tostum | Sem. Coic. Tost. |
| Corydalis Decumbentis Rhizoma | Rhiz. Coryd. Decumb. |
| Compound A-Keto Acid | A-Keto Acid Co. |
| Magnoliae Officinalis Cortex | Cort. Magn. Off. |
| Amomi Fructus Rotundus | Fruct. Amom. Rot. |
| Linagliptin | Linaglipt. |
| Entecavir | Entecav. |
| Moutan Cortex | Cort. Moutan |
| Uncariae Ramulus Cum Uncis | Ram. Unc. c. Unc. |
| Lycopi Herba | Herb. Lycop |
| Febuxostat | Febux. |
| Eucommiae Cortex | Cort. Eucomm. |
| Persicae Semen | Sem. Pers. |
| Dendrobii Caulis | Caul. Dendrob. |
| Gardeniae Fructus | Fruct. Gard. |
| Saposhnikoviae Radix | Rad. Saposhn. |
| Cassiae Semen | Sem. Cass. |
| Curcumae Rhizoma | Rhiz. Curc. |
| Dioscoreae Nipponicae Rhizoma | Rhiz. Diosc. Nipp. |
| Fructus Hordei Germinatus | Fruct. Hord. Germ. |

| | |
|---|---|
| Aconitum Carmichaelii Preparatum | Acon. Carm. Prep. |
| Carvedilol Tablets | CT |
| Citri Reticulatae Pericarpium | Pericarp. Cit. Ret. |
| Hedysari Radix | Rad. Hedys. |
| Hordei Fructus Germinatus | Fruct. Hord. Germ. |
| Angelicae Sinensis Radix | Rad. Ang. Sin. |
| Levothyroxine | Levothyrox. |
| Polygoni Multiflori Radix Praeparata | Rad. Polyg. Mult. Praep. |
| Sinapis Semen | Sem. Sinap. |
| Rosae Laevigatae Fructus | Fruct. Ros. Laevig. |
| Valsartan | Valsart. |
| Atorvastatin | Atorvast. |
| Polygoni Avicularis Herba | Herb. Polyg. Avic. |
| Eclipta Prostrata | Eclipt. |
| Epimedii Folium | Fol. Epim. |
| Spatholobi Caulis | Caul. Spath. |
| Perillae Fructus | Fruct. Perill. |
| Polygonati Rhizoma | Rhiz. Polygon. |
| Metformin Hydrochloride Tablets | MHC |
| Puerariae Lobatae Radix | Rad. Puer. Lob. |
| Dalteparin | DNS |
| Paeoniae Radix Alba | Rad. Paeon. Alb. |
| Phellodendri Chinensis Cortex | Cort. Phell. Chin. |
| Shenyan Kangfu Pian | SKP |
| Losartan Potassium | LP |
| Nelumbinis Semen | Sem. Nelumb. |
| Amomi Fructus | Fruct. Amom. |
| Astragali Radix Praeparata Cum Melle | Rad. Astrag. Praep. c. Mel. |
| Frusemide | Frusem. |
| Rehmannia Glutinosa | Rehmann. |
| Erythropoietin | EPO |
| Rabeprazole | Rabepraz. |
| Pinelliae Rhizoma Praeparatum | Rhiz. Pinell. Praep. |
| Panax Notoginseng Powder | Pulv. Panax Notog. |
| Aurantii Fructus | Fruct. Aurant. |
| Zea Mays Stigmas | Stigm. Zeae |
| Corydalis Rhizoma | Rhiz. Coryd. |
| Arisaema Cum Bile | Aris. c. Bil. |
| Cistanches Herba | Herb. Cistan. |
| Pseudostellariae Radix | Rad. Pseudostell. |
| Human Albumin | Alb. Hum. |
| Coicis Semen | Sem. Coic. |
| Arctii Fructus | Fruct. Arct. |

| | |
|---|---|
| Astragali Radix | Rad. Astrag. |
| Chrysanthemi Flos | Fl. Chrysanth. |
| Aspirin Enteric-Coated Tablets | AEC |
| Liquidambarisfructus | Fruct. Liquidamb. |
| Haliotidis Concha | Conch. Haliot. |
| Leonuri Herba | Herb. Leon. |
| Rheum Palmatum L. | Rheum palm. |
| Gastrodiae Rhizoma | Rhiz. Gastrod. |
| Valsartan And Amlodipine Tablets | Tab. Vals. & Amlod. |
| Colchicine | Colchic. |
| Schizonepetae Spica | Spic. Schizon. |
| Rehmanniae Radix Praeparata | Rad. Rehmann. Praep. |
| Smilacis Glabrae Rhizoma | Rhiz. Smilac. Glabr. |
| Fluvastatin Sodium | Fluvast. Na |
| Rhei Radix Et Rhizoma | Rad. et Rhiz. Rhei |
| Sepiae Endoconcha | Endoconch. Sep. |
| Human Immunoglobulin | Ig Hum. |
| Asteris Radix Et Rhizoma | Rad. et Rhiz. Aster |
| Plantaginis Herba | Herb. Plantag. |
| Polysaccharide Iron | Fe Polysac. |
| Ligustri Lucidi Fructus | Fruct. Ligust. Luc. |
| Arecae Pericarpium | Pericarp. Arec. |
| Trichosanthis Radix | Rad. Trichos. |
| Plantaginis Semen | Sem. Plantag. |
| Coptidis Rhizoma | Rhiz. Copt. |
| Achyranthis Bidentatae Radix | Rad. Achyr. Bident. |
| Ficus Simplicissima | Ficus simpl. |
| Cornus Officinalis With Salt | Corn. off. c. Sal. |
| Ophiopogonis Radix | Rad. Ophiopog. |
| Notoginseng Radix Et Rhizoma | Rad. et Rhiz. Notogins. |
| Armeniacae Semen Amarum | Sem. Armen. Am. |
| Atorvastatincalcium | Atorvast. Ca |
| Metformin Hydrochloride Tablets | MHC |
| Epo | EPO |
| Sodium Zirconium Cyclosilicate | Zr Cyclosil. Na |
| Corni Fructus | Fruct. Corni |
| Cirsii Herba | Herb. Cirs. |
| Metoprolol | Metoprol. |
| Rubi Fructus | Fruct. Rubi |
| Yishen Huashi Granules | YSHG |
| Menthae Haplocalycis Herba | Herb. Menth. Haploc. |
| Folic Acid Tablets | FAT |
| Torasemide Tablets | TOR |

| | |
|---|---|
| Morindae Officinalis Radix | Rad. Morind. Off. |
| Monascus Purpureus | Monasc. purp. |
| Dioscoreae Rhizoma | Rhiz. Diosc. |
| Sodium Bicarbonate | SBC |
| Iguratimod Tablets | IGU Tab. |
| Lilii Bulbus | Lil. Bulb. |
| Massae Medicatae Fermentatae Tostum | M. Medic. Ferr. Tost. |
| Gardeniae Fructus Praeparatus | Gard. Fruct. Praep. |
| Phellodendri Amurensis Cortex | Phell. Amur. Cort. |
| Ephedrae Herba | Eph. Herb. |
| Ginseng Radix et Rhizoma | Gins. Rad. Rhiz. |
| Baricitinib Tablets | Baricitinib Tab. |
| Methotrexate Tablets | MTX Tab. |
| Methylprednisolone Tablets | Methylprednisolone Tab. |
| Paridis Rhizoma | Parid. Rhiz. |
| Prednisone Acetate Tablets | PA |
| Cicadae Periostracum | Cicad. Perios. |
| KunXianJiaoNang | KXJN |
| Leflunomide Tablets | LEF |
| Tinosporae Caulis | Tinos. Caul. |
| Angelicae Pubescentis Radix | Ang. Pub. Rad. |
| Taxilli Herba | Tax. Herb. |
| Citri Reticulatae Pericarpium | Cit. Ret. Pericarp. |
| Millettiae Speciosae Radix | Mill. Spec. Rad. |
| Coriolus | Coriol. |
| Scorpio | Scorp. |
| Angelicae Dahuricae Radix | Ang. Dah. Rad. |
| Erigerontis Herba | Erig. Herb. |
| LeiGongTenDuoGanPian | LGTDGP |
| Curcumae Longae Rhizoma | Curc. Long. Rhiz. |
| Stephaniae Tetrandrae Radix | Steph. Tet. Rad. |
| Mori Ramulus | Mori Ramulus |
| Siegesbeckiae Herba | Sieg. Herb. |
| ZhengQingFengTongNingHuanShiPian | ZQFTNHSP |
| Cibotii Rhizoma | Cib. Rhiz. |
| Drynariae Rhizoma | Dryn. Rhiz. |
| Dipsaci Radix | Dip. Rad. |
| BaiShaoZongGanJiaoNang | BSZGJN |
| Cannabis Semen | Cann. Sem. |
| Kochiae Fructus | Koch. Fruct. |
| Glehniae Radix | Glehn. Rad. |
| Sophorae Flavescentis Radix | Soph. Flav. Rad. |
| Perillae Folium | Perill. Fol. |

| | |
|---|---|
| Dictamni Cortex | Dict. Cort. |
| Cynanchi Paniculati Radix et Rhizoma | Cyn. Pan. Rad. Rhiz. |
| BiQiJiaoNang | BQJN |
| Trachelospermi Caulis | Trache. Caul. |
| Poria Cum Radix Pini | Por. Cum Rad. Pin. |
| Notopterygii Rhizoma et Radix | Notop. Rhiz. Rad. |
| Glechomae Herba | Glech. Herb. |
| Recombinant Human Tumor Necrosis Factor-α Receptor II: IgG Fc Fusion Protein | TNFR-Fc |
| Aurantii Fructus Immaturus | Aur. Fruct. Imm. |
| Polygalae Radix Praeparata | Polyg. Rad. Praep. |
| Dendrobii Nobile Caulis | Dendrob. Nob. Caul. |
| Adalimumab | ADA |
| Citri Sarcodactylis Fructus | Cit. Sarc. Fruct. |
| Os Draconis | Os Drag. |
| Eriobotryae Folium | Eriob. Fol. |
| Cayratiae Herba | Cayr. Herb. |
| Benincasae Exocarpium | Ben. Exocarp. |
| Etoricoxib Tablets | ETX |
| Zaocys | Zaoc. |
| Cyclosporine Soft Capsules | CsA |
| Erythrinae Cortex | Eryth. Cort. |
| Celecoxib Soft Capsules | Celecoxib Cap. |
| Achyranthis Bidentatae Radix Praeparata Cum Vino | Achyr. Bid. Rad. Praep. Vin. |
| Piperis Kadsurae Caulis | Pip. Kad. Caul. |
| Arcae Concha Praeparata | Arc. Conch. Praep. |
| Luffae Fructus Retinervus | Luff. Fruct. Ret. |
| Cimicifugae Rhizoma | Cimic. Rhiz. |
| Artemisiae Scopariae Herba | Artem. Scop. Herb. |
| Eupatorii Herba | Eupat. Herb. |
| Crataegus Pinnatifida | Crataeg. Fruct. Tost. |
| Galli Gigerii Endothelium Corneum | Gall. Gig. Endoth. Corn. |
| Diclofenac Sodium Sustained - release Tablets | DSST |
| Tofacitinib Tablets | TFC |
| Carthami Flos | Carth. Flos |
| Ziziphi Spinosae Semen | Ziziph. Spin. Sem. |
| Albiziae Cortex | Alb. Cort. |
| Nidus Vespae | Nid. Vesp. |
| Lonicerae Caulis | Lon. Caul. |
| Hydroxychloroquine Sulfate Tablets | HCQ |
| Lycopodii Herba | Lycop. Herb. |
| WangBiJiaoNang | WBJN |

| | |
|---|---|
| Glechomae Longitubae Herba | Glech. Long. Herb. |
| Sinomenii Caulis | Sinom. Caul. |
| Phyllanthi Fructus | Phyll. Fruct. |
| Aurantii Fructus Immaturus Praeparatus Cum Braisiae | Aur. Fruct. Imm. Praep. Brais. |
| Gleditsiae Spina | Gled. Spin. |
| Crataegi Fructus | Crataeg. Fruct. |
| Massae Medicatae Fermentatae | M. Medic. Ferr. |
| Myristicae Semen Praeparatum Cum Braisiae | Myrist. Sem. Praep. Brais. |
| Psoraleae Fructus | Psoral. Fruct. |
| Momordicae Fructus | Momord. Fruct. |
| Gossampini Flos | Goss. Flos |
| Phragmitis Rhizoma | Phrag. Rhiz. |
| Mume Fructus | Mum. Fruct. |
| Compound Glycyrrhizin Tablets | CGT |
| Artemisiae Argyi Folium | Artem. Arg. Fol. |
| Micae Lapis Aureus | Mic. Lap. Aur. |
| Diammonium Glycyrrhizinate Enteric - coated Capsules | DGEC |
| Aconiti Lateralis Radix Praeparata | Acon. Lat. Rad. Praep. |
| Periplocae Cortex | Peripl. Cort. |
| Bombyx Batryticatus | Bomb. Batryt. |
| Atractylodis Macrocephalae Rhizoma Praeparatum Cum Braisiae | Atr. Macro. Rhiz. Praep. Brais. |
| Citri Grandis Exocarpium Rubrum | Cit. Grand. Exocarp. Rub. |
| Sarcandrae Herba | Sarc. Herb. |
| Iris domestica | I. domestica |
| Trichosanthes kirilowii (Cortex) | T. kirilowii |
| Commiphora myrrha (Vinegar-processed) | C. myrrha |
| Zanthoxylum bungeanum | Z. bungeanum |
| Boswellia carterii | B. carterii |
| Clematis armandii (Caulis) | C. armandii |
| Curculigo orchioides | C. orchioides |
| Prinsepia uniflora | P. uniflora |
| Eupolyphaga sinensis | E. sinensis |
| Pittosporum glabratum | P. glabratum |
| Gypsum Fibrosum | G. Fibrosum |
| Mycophenolate Mofetil Capsules | MMF |
| Coix lacryma-jobi var. ma-yuen (Bran-fried) | C. lacryma-jobi var. ma-yuen |
| Triticum aestivum var. aestivum (Blighted) | T. aestivum var. aestivum |
| Tribulus terrestris | T. terrestris |
| Oroxylum indicum | O. indicum |
| Psoralea corylifolia (Salt-processed) | P. corylifolia |

| | |
|---|---|
| Lycium chinense | L. chinense |
| Stemona sessilifolia | S. sessilifolia |
| Ephedra sinica (Honey-processed) | E. sinica |
| Aster tataricus (Honey-processed) | A. tataricus |
| Tussilago farfara | T. farfara |
| Schizonepeta tenuifolia | S. tenuifolia |
| Peucedanum praeruptorum | P. praeruptorum |
| Oryza sativa (Glutinous rice root) | O. sativa |
| Terminalia chebula | T. chebula |
| Cynanchum stauntonii | C. stauntonii |
| Diospyros kaki (Persistent calyx) | D. kaki |
| Arisaema erubescens (Processed) | A. erubescens |
| Prunus persica (Parched seed) | P. persica |
| Zanthoxylum nitidum | Z. nitidum |
| Rosa laevigata (Fructus, fleshy part) | R. laevigata |
| Endothelium Corneum Gigeriae Galli (Stir-fried) | E. Corneum |
| Scrophularia ningpoensis | S. ningpoensis |
| Cinnamomum cassia | C. cassia |
| Morus alba (Cortex) | M. alba |
| Vitex negundo var. cannabifolia (Fructus) | V. negundo |
| Ganoderma capense | G. capense |
| Polygonatum odoratum | P. odoratum |
| Acorus tatarinowii | A. tatarinowii |
| Semen Sojae Praeparatum (Fermented soybean) | S. Sojae |
| Curcuma phaeocaulis | C. phaeocaulis |
| Sparganium stoloniferum | S. stoloniferum |
| Dolichos lablab (Flos) | D. lablab |
| Magnolia officinalis (Flos) | M. officinalis |
| Myristica fragrans | M. fragrans |
| Polygonum perfoliatum | P. perfoliatum |
| Morus alba (Ramulus) | M. alba |
| Agkistrodon | A. spp. |
| Polygala tenuifolia (Honey-processed) | P. tenuifolia |
| Prunella vulgaris | P. vulgaris |
| Aconitum carmichaelii (Processed) | A. carmichaelii |
| Magnolia biondii | M. biondii |
| Turpinia arguta (Folium) | T. arguta |
| Xanthium sibiricum (Fructus) | X. sibiricum |
| Triamcinolone Acetonide Acetate Injection | TAA |
| Sterculia lychnophora | S. lychnophora |
| Saussurea involucrata | S. involucrata |
| Corydalis yanhusuo (Vinegar-processed) | C. yanhusuo |
| Panax ginseng (Folium) | P. ginseng |

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| Eriobotrya japonica (Honey-processed Folium) | E. japonica |
| Oryza sativa (Germinatus) | O. sativa |
| Polygonum chinense | P. chinense |
| Platycladus orientalis (Semen) | P. orientalis |
| Linum usitatissimum (Semen) | L. usitatissimum |
| Concha Ostreae (Praeparata) | C. Ostreae |
| Taraxacum mongolicum | T. mongolicum |
| Caesalpinia sappan (Lignum) | C. sappan |
| Perilla frutescens (Caulis) | P. frutescens |
| Dimocarpus longan (Arillus) | D. longan |
| Chimonanthus praecox (Flos) | C. praecox |
| Atractylodes macrocephala (Stir-fried) | A. macrocephala |
| Dolichos lablab (Semen) | D. lablab |
| Os Draconis Praeparata | O. Draconis |
| Atractylodes macrocephala (Stir-fried with clay) | A. macrocephala |
| Cervus nippon (Cornu, powdered) | C. nippon |
| Phellodendron amurense (Salt-processed) | P. amurense |
| Desmodium styracifolium | D. styracifolium |
| Tetrapanax papyrifer (Medulla) | T. papyrifer |
| Agrimonia pilosa | A. pilosa |
| Sanguisorba officinalis (Carbonised) | S. officinalis |
| Ziziphus jujuba var. spinosa (Stir-fried Semen) | Z. jujuba |
| Nelumbo nucifera (Folium) | N. nucifera |
| Morus alba (Folium) | M. alba |
| Dicliptera chinensis | D. chinensis |
| Vitex trifolia (Fructus) | V. trifolia |
| Cynomorium songaricum | C. songaricum |
| Curcuma aromatica | C. aromatica |
| Santalum album | S. album |
| Ligusticum sinense | L. sinense |
| Citrus aurantium (Bran-fried Exocarpium) | C. aurantium |
| Chinemys reevesii (Vinegar-processed) | C. reevesii |
| Golimumab | GLM |
| Stauntonia chinensis | S. chinensis |
| Scolopendra subspinipes | S. subspinipes |
| Bombyx mori Faeces | B. mori |
| Sulfasalazine Enteric-coated Tablets | SSZ |
| Asparagus cochinchinensis | A. cochinchinensis |
| Zhengqing Fengtongning Injection | ZQFTN |
| Inula japonica | I. japonica |
| Haematitum | H. spp. |
| Arca subcrenata | A. subcrenata |
| Trichosanthes kirilowii (Semen) | T. kirilowii |

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|---|---|
| Moghania philippinensis | M. philippinensis |
| Phellodendron amurense (Stir-fried) | P. amurense |
| Chinemys reevesii (Carapax) | C. reevesii |
| Etanercept Injection | ETN |
| Alpinia oxyphylla | A. oxyphylla |
| Polysaccharide-Iron Complex Capsules | PIC |
| Sanguisorba officinalis | S. officinalis |
| Platycladus orientalis | P. orientalis |
| Lindera aggregata | L. aggregata |
| Raphanus sativus (Tostus) | R. sativus |
| Luobitong Pian | L. Bitong |
| Bambusae concretio silicea | B. concretio |
| Croton crassifolius | C. crassifolius |
| Semiliquidambar cathayensis | S. cathayensis |
| Ardisia gigantifolia | A. gigantifolia |
| Pinellia ternata (Praeparata) | P. ternata |
| Acorus tatarinowii | A. tatarinowii |
| Fagopyrum dibotrys | F. dibotrys |
| Pregabalin Capsules | PGB |
| Cremastra appendiculata | C. appendiculata |
| Ursodeoxycholic Acid Capsules | UDCA |
| Methylprednisolone | MP |
| Prednisone | PDN |
| Cyclosporine Capsules | CYC |
| Metoprolol | MET |
| Alfacalcidol | ALF |
| Scutellaria barbata | S. barbata |
| Lithospermum erythrorhizon | L. erythrorhizon |
| Valsartan Capsules | VAL |
| Juncus effusus | J. effusus |
| Hydrochlorothiazide | HCTZ |
| Mel | Mel |
| Sophora japonica | S. japonica |
| Isatis indigotica | I. indigotica |
| Panax quinquefolius | P. quinquefolius |
| Trionyx sinensis (Praeparata) | T. sinensis |
| Typha angustifolia | T. angustifolia |
| Pulsatilla chinensis | P. chinensis |
| Dioscorea hypoglauca | D. hypoglauca |
| Magnetitum | Magnetitum |
| Vigna umbellata | V. umbellata |
| Cirsium japonicum | C. japonicum |
| Pteris multifida | P. multifida |

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|---|---|
| Pumex | Pumex |
| Euonymus alatus | E. alatus |
| Sargassum fusiforme | S. fusiforme |
| Citrus reticulata (Semen) | C. reticulata |
| Oryza sativa | O. sativa |
| Tripterygium hypoglaucum | T. hypoglaucum |
| Ganoderma lucidum | G. lucidum |
| Litchi chinensis (Semen) | L. chinensis |
| Nelumbo nucifera (Plumula) | N. nucifera |
| Pyrola calliantha | P. calliantha |
| Fossilia Dentis Mastodi | F. Dentis |
| Rosa rugosa | R. rugosa |
| Lasiosphaera fenzlii | L. fenzlii |
| Nelumbo nucifera (Nodus) | N. nucifera |
| Bubali Cornu | B. Cornu |
| Punica granatum (Pericarpium) | P. granatum |
| Massa Medicata Fermentata | M. Fermentata |
| Brainea insignis | B. insignis |
| Cnidium monnieri | C. monnieri |
| Viola philippica | V. philippica |
| Areca catechu | A. catechu |
| Bletilla striata | B. striata |
| Rehmannia glutinosa | R. glutinosa |
| Ilex asprella | I. asprella |
| Trogopterus xanthipes (Faeces) | T. xanthipes |
| Allium macrostemon | A. macrostemon |
| Alpinia oxyphylla | A. oxyphylla |
| Mycophenolate Mofetil | MMF |
| Rabeprazole Sodium Enteric-coated Capsules | RAB |
| Allisartan Isoproxil | ALS |
| Insulin Lispro Protamine Mixed Suspension | ILP |
| Belimumab | BEL |
| Metformin | MET |
| Azathioprine | AZA |
| Niaodukang Heji | NDK |
| Repaglinide | RPG |
| Felodipine | FEL |
| Aconitum carmichaelii (Praeparata) | A. carmichaelii |
| Clopidogrel | CLO |
| Bailing Jiaonang | B. Jiaonang |
| Calcium Dobesilate | CaD |
| Nifedipine | NIF |
| Vitamin B Complex | VBC |

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|---|---|
| Amlodipine | AML |
| Alendronate Sodium | ALE |
| Liniu Heji | L. Heji |
| Ezetimibe | EZE |
| Dapagliflozin | DAPA |
| Olmesartan Medoxomil | OLM |
| Fenofibrate | FEN |
| Dulaglutide | DULA |
| Lanthanum Carbonate | LAN |
| Tolvaptan | TOLV |
| Finerenone | FIN |
| Sacubitril/Valsartan Sodium | SAC/VAL |
| Canagliflozin | CANA |
| Insulin Lispro Protamine | ILP |
| Cinacalcet | CIN |
| Losartan Potassium/Hydrochlorothiazide | LOS/HCTZ |
| Callicarpa nudiflora | C. nudiflora |
| Human Albumin (20%) | HA |
| Esomeprazole | EZO |
| Benazepril | BZP |
| Irbesartan Tablets | IRB |
| Irbesartan and Hydrochlorothiazide Tablets | IRB/HCTZ |
| Empagliflozin Tablets | EMPA |
| Entecavir Tablets | ETV |
| Febuxostat Tablets | FEB |
| Fluvastatin | FLV |
| Calcium Tablets | Calcium Tablets |
| Heparin Sodium | HP |
| Pueraria lobata (Willd.) Ohwi | P. lobata |
| Metoprolol Tartrate | MTP |
| Canagliflozin Acetate Tablets | CANA |
| Canagliflozin Tablets | CANA |
| Entecavir Maleate Tablets | ETV |
| Calcium Carbonate and Vitamin D3 Tablets | CCDD3 |
| Lepidium apetalum Willd. | TOR |
| Benazepril Hydrochloride Tablets | BZP |
| Prunus armeniaca L. | P. armeniaca |
| Benazepril Hydrochloride Tablets | BZP |
| YiLing KouFuYe | YLKFY |
| Cordyceps Militaris Mycelium Powder Capsule | CMPC |
| Zhenqi Fuzheng Tablet | ZQFZT |